Lab Report 1

Course Code: CSE-3232

Course Title: Computer Network Sessional

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Title:

Establishing and setting up a network with ten computers and two hubs, then sending messages from one computer to another to check for connectivity.

Objective:

- Use Cisco Packet Tracer to implement a peer-to-peer network.
- How a network can be established using a hub.
- Using numerous hubs to link two networks together.

Background Theory:

A hub is a fundamental networking device used to connect multiple computers or devices within a network. It functions at the physical layer of the OSI model and lacks the ability to intelligently direct data. Instead, when a device sends data through a hub, the hub simply broadcasts the data to all connected devices, regardless of the intended recipient. Only the device with the matching address processes the data, while all others discard it.

Observation and Findings:

During the simulation, I connected 10 PCs using two hubs in Packet Tracer. Each hub was connected to 5 PCs, which made sense given that each hub had only 6 ports. The simulation picture is given below:

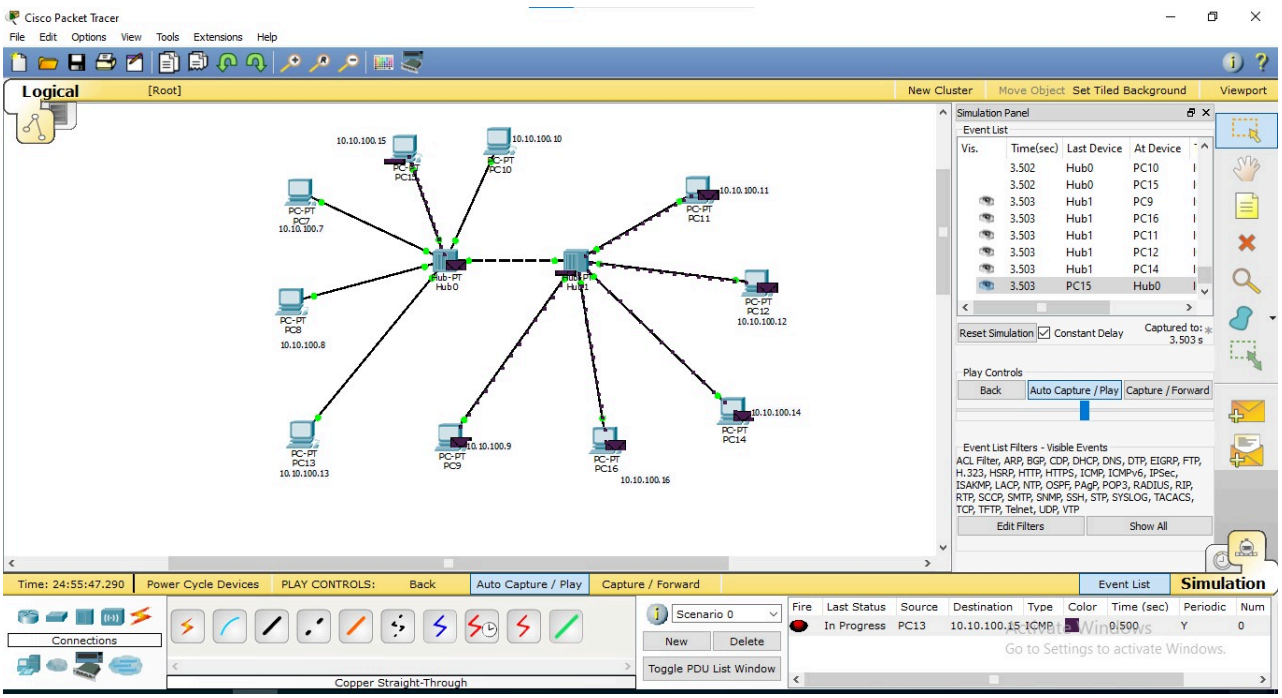

Figure: Hub-based network of ten computers.

Some key observations were:

- **Limitations of Hub Port**: We had to use two hubs because each hub only had six ports, which was insufficient to link all ten PCs. This demonstrates how hubs can be constrictive when additional devices need to be connected.
- **Data Broadcasting**: All of the linked devices received data broadcast by the hub whenever one PC sent it. As anticipated, however, the message was disregarded by the other PCs and only processed by the destination PC. This demonstrated that hubs are unable to target particular devices. All they do is send out data.
- **Network Behaviors**: As is common in a hub-based network, only the PC for whom the message was meant reacted, even if all PCs got the data. The others simply throw it out, generating needless traffic.

Discussion:

The experiment highlights the limitations of hubs in a network. Hubs have limited ports, requiring multiple hubs for larger networks, making the setup complex. They inefficiently use bandwidth by broadcasting data to all devices, leading to network congestion. Additionally, hubs operate in a shared environment, causing data collisions that slow down communication.

Conclusion:

Hubs are suitable for small networks but inefficient for larger ones. Their broadcasting nature and limited scalability cause congestion and slowdowns. For better performance, switches or routers should be used in larger networks to handle traffic efficiently.